

CSF beta-amyloid levels are altered in narcolepsy: a link with the inflammatory hypothesis?

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SUMMARY

Narcolepsy is characterized by hypocretin deficiency due to the loss of hypothalamic orexinergic neurons, and is associated with both the human leucocyte antigen DQB1*06:02 and the T cell receptor polymorphism. The above relationship suggests autoimmune/inflammatory processes underlying the loss of orexinergic neurons in narcolepsy. To test the autoimmune/inflammatory hypothesis by means of cerebrospinal fluid (CSF) levels of beta-amyloid_{1–42} and/or total tau proteins in a sample of narcoleptic patients, we analysed 16 narcoleptic patients and 16 healthy controls. Beta-amyloid_{1–42} CSF levels were significantly lower in narcoleptic patients compared with healthy controls. We also documented pathologically low levels of CSF beta-amyloid_{1–42} (<500 pg mL^{−1}) in six of 16 narcoleptic patients (37.5%). We hypothesize that the significant decrease of the CSF beta-amyloid_{1–42} levels in narcoleptic patients may support both the inflammatory/autoimmune hypothesis as the basis of the pathogenesis of narcolepsy and the prevalence of an 'amyloidogenic' pathway caused by the deficiency of the alpha-secretases enzymes.

INTRODUCTION

Narcolepsy is associated with hypocretin/orexin-A (orx) deficiency due to the dramatic loss of hypothalamic orexinergic neurons, especially in narcolepsy with cataplexy (NC) (Nishino *et al.*, 2000; Thannickal *et al.*, 2000). Narcolepsy with orx deficiency is associated strongly with the human leucocyte antigen (HLA) DQB1*06:02 and the T cell receptor polymorphism (Fontana *et al.*, 2010). The above link suggests that unknown autoimmune/inflammatory processes may underlie the loss of orexinergic neurons (Cvetkovic-Lopes *et al.*, 2010; Korum *et al.*, 2011). Upper airway *Streptococcus pyogenes* infections and influenza-A H1N1 infections and vaccinations have been considered as possible autoimmune triggers in genetically predisposed subjects (Korum *et al.*, 2011). Recently, low cerebrospinal fluid (CSF) beta-amyloid_{1–42} (A β _{1–42}) levels have been described in a case of NC post-H1N1 vaccination and also in an unpublished case of NC lacking previous history of vaccination/infection (Kallweit *et al.*, 2012). It has been hypothesized that the low A β _{1–42} CSF levels encountered in genetically

recognizable patients could be related to post-infectious immunological processes.

The aim of our study was to evaluate the CSF levels of A β _{1–42}, total-tau (t-tau) and phosphorylated-tau (p-tau) in narcoleptic patients, compared with matched healthy controls, to test the hypothesis of an inflammatory pathway in narcolepsy interfering with CSF A β _{1–42} and/or t-tau and p-tau metabolisms.

SUBJECTS AND METHODS

We included 16 consecutive drug-naïve narcoleptic inpatients (mean age 36.12 ± 10.78 years; age range 18–62 years; seven men, nine women) referred to the sleep centre of the University Hospital 'Tor Vergata' in Rome (Italy), diagnosed as affected by narcolepsy without cataplexy (NwC) and NC according to the *International Classification of Sleep Disorders*, 2nd edition (2005). All patients underwent a diagnostic/study protocol including clinical history collection, neurological examination and laboratory polysomnography, followed by the Multiple Sleep Latency

Test (MSLT), brain magnetic resonance imaging (MRI) scan, HLA DQB1*06:02 haplotype determination and lumbar puncture to detect orx, $A\beta_{1-42}$, t-tau and p-tau CSF levels. Sixteen healthy controls (HC) (mean age 33.50 ± 11.84 years; age range 18–63 years; seven men, nine women), similar for age ($P = 0.82$) and sex-matched with narcoleptic patients, were recruited prospectively at the same university hospital. All HC underwent lumbar puncture for diagnostic purposes, 14 for suspected multiple sclerosis and two for suspected subarachnoid haemorrhage. At the end of the diagnostic study the clinical and instrumental data excluded CNS or systemic disorders in HC. Exclusion criteria for narcoleptic patients and HC were intake of drugs interfering with CNS, inflammatory/infectious conditions and anti-inflammatory or corticosteroid therapies. Peripheral blood and CSF were collected from patients and controls. Routine determinations included total cell count, measurement of the concentration of total proteins and albumin, both in CSF and serum, and immunoglobulin IgG index (as indicator of intrathecal IgG production). The presence of oligoclonal bands (OB) was investigated in both narcoleptic patients and controls. CSF withdrawal was performed in patients and controls between 08:00 and 09:00 A.M. using an atraumatic needle. After collection, CSF was centrifuged immediately to eliminate cells and cellular debris and stored at -80°C until analysed. Participants were informed about the study purpose and procedures. Written informed consent was obtained from all the subjects. The local Ethical Committee of the University Hospital of Rome 'Tor Vergata' gave their approval to the study.

CSF determination of $A\beta_{1-42}$, t-tau, p-tau and orx levels

The levels of $A\beta_{1-42}$, t-tau and p-tau were determined according to previously published standard procedures (Sancesario *et al.*, 2010) using commercially available sandwich enzyme-linked immunosorbent assays (ELISA) (Innotest β -amyloid 1–42, Innotest h-T-tau, Innotest Phospho-T-tau 181; Innogenetics, Ghent, Belgium).

The orx CSF levels were detected with a commercially available ELISA kit (Orexin A/Hypocretin-1 EIA Kit; Phoenix Pharmaceuticals, Burlingame, CA, USA), based on the principle of competitive enzyme immunoassay. One hundred μL of CSF samples or standard peptides were dispensed in duplicate and incubated together with 50 μL of biotinylated orexin-peptides and with 50 μL of primary antibody, which binds to secondary antibody precoated on the plate. After washing, the biotinylated peptides were detected by peroxidase-labelled streptavidin horseradish peroxidase (SA-HRP), after adding a substrate solution. The reaction was stopped by hydrochloric acid and the absorbance was read at 450 nm. The intensity of the colorimetric reaction was directly proportional to the amount of the biotinylated peptide–SA-HRP complexes and inversely proportional to the amount of peptide standards or samples. This is due to the competitive binding of

biotinylated peptides, standard peptides or samples to the primary antibody. The unknown sample concentrations were then calculated on the corresponding standard sigmoid curve equation (peptide standards ranging from 0 to 1000 pg mL^{-1}).

Data analysis

Statistical analysis was performed by means of the non-parametric Mann–Whitney *U*-test (Statistica version 10.0; Statsoft Inc., Tulsa, OK, USA). A *P*-value <0.05 was considered significant. Finally, Spearman's correlation coefficients were calculated as and when appropriate.

RESULTS

Subjects

We collected CSF from 16 patients (nine NwC and seven NC), and from 16 HC (see Tables 1 and 2). In both narcoleptic patients and HC, OB were not detected. Moreover, the IgG index was normal ($<4 \text{ mg dL}^{-1}$) except in a NC patient, who displayed an increased IgG index (18.1 mg dL^{-1}). This patient also reported narcolepsy onset some weeks after influenza infection in 2009; he was HLA DQB1*06:02 negative (see Table 2, NC patient no. 7). Another patient showed a narcoleptic symptom onset after a mild traumatic brain injury (TBI). He was affected by NwC, and showed 81.4 pg mL^{-1} orx and $369 A\beta_{1-42}$ CSF levels. Brain MRI scans were unremarkable in all narcoleptic patients.

Table 1 CSF data of narcoleptic patients and controls

Demographic data	Narcoleptic patients (<i>n</i> = 16) (mean $\pm$ SD)	Controls (<i>n</i> = 16) (mean $\pm$ SD)	<i>P</i> value
Age	36.12 $\pm$ 10.78	33.5 $\pm$ 11.84	0.82
Gender	7M 9F	7M 9F	NA
Disease duration (years)	10.96 $\pm$ 12.18	NA	NA
CSF data (expressed in pg mL^{-1})	Narcoleptic patients (<i>n</i> = 16) (mean $\pm$ SD)	Controls (<i>n</i> = 16) (mean $\pm$ SD)	<i>P</i> value
Orexin A	77.89 $\pm$ 37.72	149.38 $\pm$ 30.79	0.00004*
t-tau Protein	184.37 $\pm$ 76.17	219.06 $\pm$ 62.74	0.21
p-tau Protein	32.31 $\pm$ 10.77	37.19 $\pm$ 10.2	0.16
Beta Amyloid ₁₋₄₂	656 $\pm$ 266.43	1068.44 $\pm$ 143.39	0.00019*

*Significance level $P < 0.05$.

Table 2 Clinical, polysomnographic and CSF features of narcoleptic patients

Diagnosis	HLA	MSL (min)	SOREMPs (no.)	Orx	A β_{1-42}	t-tau	p-tau
NC	+	4	2	26.5	624	117	26
NC	+	6	2	52.4	725	157	26
NC	+	3.2	5	51.2	571	262	35
NC	+	4.4	2	65.8	274	214	34
NC	+	5	3	56	764	86	23
NC	+	1.25	3	28.3	407	238	24
NC	–	3.5	3	30.2	496	282	45
NwC	+	1.7	3	85	453	100	20
NwC	+	2.9	2	81.4	369	75	18
NwC	+	3.9	3	81.4	1026	144	36
NwC	+	3.6	2	124.38	1175	154	41
NwC	–	5	3	82.9	365	114	18
NwC	–	2.2	3	88.5	828	212	33
NwC	–	4.5	2	102.3	1021	254	42
NwC	–	4.6	2	143.5	812	324	55
NwC	–	5.2	3	146.5	586	217	41

NC, narcolepsy with cataplexy; NwC, narcolepsy without cataplexy; HLA, HLA DQB1 0602 haplotype; SOREMPs, sleep-onset REM periods; A β_{1-42} , beta-amyloid 1-42 CSF levels (pg mL⁻¹); t-tau, total-tau proteins CSF levels (pg mL⁻¹); p-tau, phosphorylated-tau proteins CSF levels (pg mL⁻¹); orx, Orexin-A CSF levels (pg mL⁻¹).

CSF contents of orexin-A in narcoleptic patients

CSF orx levels were significantly lower in the narcoleptic group than in the HC ($P = 0.0004$). NC ($n = 7$) showed orx mean values significantly lower than controls, and specifically lower than one-third of the mean value of the HC (NC group: 44.34 ± 15.73 ; HC: 149.38 ± 30.79 , $P = 0.0002$). NwC ($n = 9$) patients also presented orx levels significantly lower than controls ($P = 0.0027$). Furthermore, NC showed CSF orx values lower than NwC ($P = 0.001$). Data are summarized in Tables 1 and 3.

CSF contents of amyloid- β_{1-42} , t-tau and p-tau proteins in narcoleptic patients

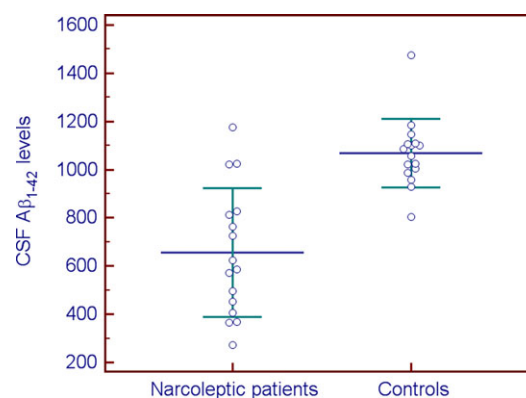
A more than one-and-a-half A β_{1-42} CSF levels decrease was detected in narcoleptic patients compared with HC ($P = 0.00019$; Table 1, Fig. 1). Six of the 16 (37.5%) narcoleptic patients showed CSF A β_{1-42} values lower than 500 pg mL⁻¹. Fifteen of the 16 narcoleptic subjects (93.75%) presented CSF A β_{1-42} lower levels than the CSF A β_{1-42} mean value of the matched controls.

Conversely, CSF levels of both t-tau ($P = 0.21$) and p-tau proteins ($P = 0.16$) did not differ statistically between groups (Table 1). CSF A β_{1-42} , t-tau and p-tau proteins levels did not diverge statistically between NwC ($n = 9$) and NC ($n = 7$) patients as well as HLA+ ($n = 10$) and HLA- ($n = 6$) subjects (Table 3).

Table 3 Comparing CSF and PSG data between narcolepsy with cataplexy (NC) and narcolepsy without cataplexy (NwC) patients, and narcoleptic patients with (HLA+) or without DQB1*06:02 polymorphism (HLA-)

CSF data (expressed in pg mL ⁻¹)	NC patients (n = 7) (mean $\pm$ SD)	NwC patients (n = 9) (mean $\pm$ SD)	P value
Orexin A	44.34 $\pm$ 15.73	103.99 $\pm$ 27.05	0.001*
t-tau Protein	179 $\pm$ 70.1	177.11 $\pm$ 80.84	0.52
p-tau Protein	30.42 $\pm$ 7.98	33.78 $\pm$ 12.82	0.75
Beta	551.57 $\pm$ 174.1	737.22 $\pm$ 305.61	0.24
Amyloid ₁₋₄₂			
PSG data			
MSL (min)	3.91 $\pm$ 1.5	3.7 $\pm$ 1.24	0.92
SOREMPs (n°)	2.86 $\pm$ 1.07	2.55 $\pm$ 0.53	0.72
CSF data (expressed in pg mL ⁻¹)	HLA+ (n = 10) (mean $\pm$ SD)	HLA- (n = 6) (mean $\pm$ SD)	P value
Orexin A	65.2 $\pm$ 29.2	98.98 $\pm$ 43.24	0.0575
t-tau Protein	154.7 $\pm$ 64.6	233.83 $\pm$ 72.08	0.073
p-tau Protein	28.3 $\pm$ 7.6	39 $\pm$ 12.51	0.093
Beta	638.8 $\pm$ 290.5	684.67 $\pm$ 243.88	0.704
Amyloid ₁₋₄₂			
PSG Data			
MSL (min)	3.59 $\pm$ 1.4	4.17 $\pm$ 1.13	0.30
SOREMPs (n°)	2.7 $\pm$ 0.95	2.67 $\pm$ 0.52	0.71

*Significance level $P < 0.05$.

**Figure 1.** Cerebrospinal fluid (CSF) beta amyloid₁₋₄₂ (A β_{1-42}) levels in narcoleptic patients and controls (expressed in pg mL⁻¹).

Finally, Spearman's correlation test failed to find any correlation between CSF orx and A β_{1-42} levels in the narcoleptic group ($P = 0.15$), in NC ($P = 0.93$), in NwC ($P = 0.27$), in HLA+ ($P = 0.21$) and in HLA- ($P = 0.39$). Moreover, a correlation between A β_{1-42} CSF levels and disease duration ($P = 0.12$) was not evident.

DISCUSSION

The main result of the present study is the detection of a significant reduction in CSF levels of $A\beta_{1-42}$ in narcoleptic patients when compared with controls. It is worth noting that more than one-third (six of 16) of the narcoleptic sample showed pathological CSF values of $A\beta_{1-42}$ (<500 pg mL⁻¹), and more than 90% of patients (15 of 16) exhibited CSF $A\beta_{1-42}$ levels lower than HC mean value.

Recently, in two NC patients low CSF $A\beta_{1-42}$ levels have been described (Kallweit *et al.*, 2012).

It is known that the beta amyloid metabolism is regulated strongly by inflammatory mechanisms. Indeed, the inflammatory pathways of the immune system represent targets for modulating beta amyloid generation and accumulation (Hickman *et al.*, 2008). For instance, beta amyloid deposition could be triggered by stimulation of the innate immune system (Soscia *et al.*, 2010). Conversely, in animal models, perturbations in both orx signalling (typically disturbed in narcoleptic patients) and the sleep–wake cycle seem to have acute effects upon beta amyloid dynamics (Kang *et al.*, 2009). Although the relationship between the beta amyloid pathology and the orx disturbance has been reported recently *in vivo* in a small sample of Alzheimer's disease patients (Slats *et al.*, 2012), clarification is needed as to how CSF orx and beta amyloid levels may influence each other. Noticeably, beta amyloid metabolism was not investigated in animal models of narcolepsy.

Although several studies support an autoimmune–inflammatory pathogenesis of narcolepsy, the autoimmune theory is still considered weak (Kornum *et al.*, 2011). However, our findings may be considered as confirming the inflammatory pathogenesis of the disease, as narcolepsy is a neurological disorder caused by unclear autoimmune processes responsible for the selective loss of orexinergic neurons, due to inflammatory mechanisms that are more probably mediated by the activation of T cells (Kornum *et al.*, 2011). Notably, infections appear as the main trigger factors of autoimmune CNS disorders (Liguori *et al.*, 2013), as T cell-mediated responses produce viral and bacterial clearance but may cause severe immunopathologies. Therefore, we hypothesize that in narcoleptic patients altered beta amyloid metabolism may be due to inflammatory pathways involved in the pathogenesis of the disease.

In addition, NC patients showed a trend towards a lower decrease of CSF $A\beta_{1-42}$ than in NwC patients, probably lacking statistical significance because of the small sample of the two subgroups. As expected, NC also presented statistically lower CSF orx levels than NwC. Moreover, using the ELISA test, the NC patients showed CSF orx levels lower than one-third of the mean value of the controls, in accordance with ICSD-2 criteria (AASM, 2005). NC is considered to be linked to a more pronounced orexinergic neuron loss with lower CSF orx levels and to the strong association with the HLA haplotype DQB1*06:02 (Burgess and Scammell, 2012). These features may suggest a more

severe inflammatory condition in NC than in NwC, as advocated by the lower CSF $A\beta_{1-42}$ levels in NC patients. In narcoleptic patients, CSF OB were not present and the IgG index was within the range of normality, except for one patient with probable post-infectious narcolepsy (Fredrikson *et al.*, 1990). It might be interesting to evaluate possible inflammatory or traumatic forms of narcolepsy, even if the long disease duration added difficulty in achieving a precise clinical history regarding disease onset in our patients. Therefore we can only hypothesize that, among our narcoleptic patients, a case of NC occurred after unspecified influenza infection in 2009 (period of the influenza-A H1N1 pandemic) and another case of NwC following TBI (Poryazova *et al.*, 2011).

Interestingly, abnormally low levels of acetyl-L-carnitine (ALC) have been observed recently in narcolepsy, suggesting the dysfunction of the fatty acid β -oxidation pathways (Miyagawa *et al.*, 2011). ALC stimulates α -secretase activity such as ADAM10, which promotes the 'non-amyloidogenic' pathways preventing the formation of intact beta amyloid peptides (De Strooper *et al.*, 2010). Indeed, the reduced levels of ALC in narcoleptic patients may promote the 'amyloidogenic' pathways and subsequently the decrement in the $A\beta_{1-42}$ CSF levels, as observed in our study (Miyagawa *et al.*, 2013). However, so far there is no clear evidence of amyloidogenic plaques in a small sample of narcoleptic brains (Honda *et al.*, 2009).

Hence, at this stage, the pathogenesis of the reduced $A\beta_{1-42}$ CSF levels in narcolepsy may only be hypothesized. Although the inflammatory hypothesis is less conjectural in explaining low $A\beta_{1-42}$ CSF levels, the 'amyloidogenic' hypothesis may be not completely excluded. Multiple factors may contribute to the development of narcolepsy, which also include immune system and environmental factors (Kornum *et al.*, 2011).

Although our study does not allow generalization due to the small sample size, we hypothesize that the significant decrease of $A\beta_{1-42}$ CSF levels in narcoleptic patients may strengthen the inflammatory/autoimmune pathogenic hypothesis of narcolepsy. However, further studies are needed to clarify more effectively the role of the beta amyloid metabolism and its links with orexinergic neuron degeneration and loss.

AUTHOR CONTRIBUTIONS

CL created the study concept and design, data acquisition, analysis and interpretation and contributed to critical revision of the manuscript for important intellectual content. FP contributed to analysis and interpretation and to study supervision. MA contributed to the data acquisition, analysis and interpretation; MN contributed to the data acquisition; FI critically revised the manuscript for important intellectual content; MGM contributed to the study supervision; NBM contributed to the study supervision; SB contributed to the data acquisition; AR created the study concept and design, data acquisition, statistical analysis, analysis and interpretation and contributed to critical revision of the manuscript for important intellectual content.

CONFLICT OF INTEREST

No conflicts of interest declared.

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